
Tenacity Tutoring

Quadratics Practice Paper

Year 10 maths

Total Marks: 10 | Time Allowed: 30 minutes

Name: Sample Student Date: _____

Section A - Short answer

1. Solve $x^2 - 5x + 6 = 0$ by factorising.

[3 marks]

3. Simplify $\frac{3}{4} + \frac{2}{5}$ and express $\sqrt{50}$ in simplest surd form.

[2 marks]

Section B - Extended response

2. A rectangle has length $(2x + 3)$ cm and width $(x - 1)$ cm.

(a) Write an expression for the area in expanded form.

[2 marks]

(b) Given the area is 20 cm^2 , show this leads to $2x^2 + x - 23 = 0$.

[2 marks]

(c) Solve using $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, to 2 d.p.

[1 mark]

Mark Scheme		
Q#	Answer	Marks
Q1	$(x - 2)(x - 3) = 0$, so $x = 2$ or $x = 3$	3
Q3	$\frac{3}{4} + \frac{2}{5} = \frac{23}{20}$; $\sqrt{50} = 5\sqrt{2}$	2
Q2(a)	$2x^2 + x - 3$	2
Q2(b)	$(2x+3)(x-1) = 20$ gives $2x^2 + x - 23 = 0$	2
Q2(c)	$x = \frac{-1 \pm \sqrt{185}}{4} \approx 2.65$ or -3.15	1